$$CO + C_2H_4 + H_2 \longrightarrow CH_3CH_2CHO$$

Fischer Tropsch synthesis:

The reaction of carbon monoxide with hydrogen at a pressure of less than 50 atm using metal catalysts at 500 - 700 K yields saturated and unsaturated hydrocarbons.

$$nCO + (2n+1)H_2 \longrightarrow C_nH_{(2n+2)} + nH_2O$$

$$nCO + 2nH_2 \longrightarrow C_nH_{2n} + nH_2O$$

Carbon monoxide forms numerous complex compounds with transition metals in which the transition meal is in zero oxidation state. These compounds are obtained by heating the metal with carbon monoxide.

Eg. Nickel tetracarbonyl $[Ni(CO)_4]$, Iron pentacarbonyl $[Fe(CO)_5]$, Chromium hexacarbonyl $[Cr(CO)_6]$.

Structure:

It has a linear structure. In carbon monoxide, three electron pairs are shared between carbon and oxygen. The bonding can be explained using molecular orbital theory as discussed in XI standard. The C-O bond distance is 1.128Å. The structure can be considered as the resonance hybrid of the following two canonical forms.

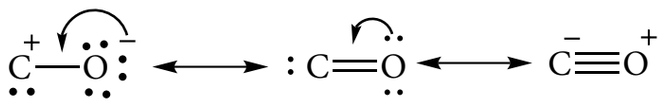


Figure 2.9 Structure of carbon monoxide

Uses of carbon monoxide:

1. Equimolar mixture of hydrogen and carbon monoxide - water gas and the mixture of carbon monoxide and nitrogen - producer gas are important industrial fuels
2. Carbon monoxide is a good reducing agent and can reduce many metal oxides to metals.
3. Carbon monoixde is an important ligand and forms carbonyl compound with transition metals

2.3.6 Carbon dioxide:

Carbon dioxide occurs in nature in free state as well as in the combined state. It is a constituent of air (0.03%). It occurs in rock as calcium carbonate and magnesium carbonate.

Production

On industrial scale it is produced by burning coke in excess of air.

$$C + O_2 \longrightarrow CO_2 \quad \Delta H = -394 \text{ kJ mol}^{-1}$$

Calcination of lime produces carbon dioxide as by product.

$$CaCO_3 \longrightarrow CaO + CO_2$$

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